

ROBOTIC COMMON SENSE AS EXECUTABLE AFFORDANCE TESTS

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ABSTRACT

Robotic “common sense” is often treated as a prior over plausible facts: mugs hold liquid, knives cut, boxes support objects. That framing is fragile for embodied agents because the dangerous cases are precisely those in which a name or visual category remains familiar while the physical variables have changed: a mug has a hole, a knife is dull, a box is wet, or the robot gripper cannot apply the required force. We propose Executable Affordance Test Logic (EATL), in which a common-sense affordance predicate is not a text fact, a static label, or a planner precondition, but a small robot-specific test program that returns a witness or failure certificate for a given object, action, robot, and task demand. We give a simple lower bound showing that any label-only prior fails on label-preserving affordance flips, and we provide a runnable synthetic tabletop stress test covering containment, lifting, load support, cutting, wiping, and hot carrying. In the flip regime, a name prior reaches 0.834 accuracy with 0.134 unsafe false positives, while EATL reaches 0.972 accuracy with 0.022 unsafe false positives at mean test cost 0.095. The evidence is not a real-robot result; it is an executable demonstration that the common field assumption—object names track physical affordances—is a mechanism-level failure point.

1 INTRODUCTION

Embodied agents inherit a tempting shortcut from language: if the object is called a mug, use mug common sense. This shortcut appears throughout robot systems that map language, object categories, or visual detections into affordance scores, skill choices, or planning preconditions. It is also often useful. A robot should not relearn from scratch that bowls usually contain liquid or that sponges usually wipe spills.

The problem is that common sense for robots is not only about what is likely. It is about what is safe to do now. A name can remain unchanged while the physical predicate changes: a cup-shaped sieve leaks, a mug has a hairline crack, a cardboard box collapses under a heavy object, a knife is decorative and dull, or a plastic tray deforms under heat. These are not exotic philosophical cases. They are ordinary deployment failures caused by hidden state, materials, wear, counterfeits, task demand, and embodiment.

This paper argues that robotic common sense should be operational rather than textual. We define a common-sense affordance claim such as `can-contain-liquid(object, 200ml)` by the executable micro-experiment that would make it true for this robot: pour a safe probe volume, observe leakage and stability, and return a witness. Similar tests can probe lift, support, cut, wipe, or carry-hot predicates. The central object is therefore not a larger language model, a better dataset, or a verifier bolted onto a plan. It is a semantics for embodied common-sense predicates.

Our contributions are:

- We formalize Executable Affordance Test Logic (EATL): affordance predicates are typed test programs over robot, object, action, demand, and observation.
- We prove a simple impossibility result for label-only common-sense priors under label-preserving affordance flips.

- We release a runnable synthetic stress test showing that the broken assumption matters under controlled physical mutations.
- We map the novelty boundary against affordance learning, task-and-motion planning, language-conditioned robotics, active sensing, and world models.

2 RELATED WORK AND NOVELTY BOUNDARY

Affordances. Affordances originate as action possibilities relative to an actor (Gibson, 1979); design work distinguishes perceived affordances and constraints (Norman, 1988). Robotics has long treated affordances as relations among entities, actions, and effects (Sahin et al., 2007; Montesano et al., 2008; Stoytchev, 2005). Interactive object learning and object-action complexes similarly ground perception in action (Fitzpatrick et al., 2003; Kruger et al., 2011). These works make generic affordance prediction non-novel. Our distinction is that an affordance predicate is accepted as robot common sense only when its executable test has a current witness.

Visual and learned affordance detection. RGB-D activity models, part-based tool affordances, and deep affordance detectors predict likely functions from observations (Koppula et al., 2013; Myers et al., 2015; Do et al., 2018). Knowledge representations encode object affordances for reasoning (Zhu et al., 2014). These methods can provide candidates or priors for EATL, but the proposed semantics rejects a prediction that lacks a task-demand-specific test witness.

Planning and feasibility. Task-and-motion planning integrates symbolic choices with geometric or continuous feasibility checks (Kaelbling & Lozano-Perez, 2011; Toussaint, 2015; Garrett et al., 2020). This makes feasibility checking inside a planner non-novel. EATL differs in scope: it defines embodied common-sense claims before plan execution and treats failure certificates as first-class outputs for hidden physical variables such as holes, porosity, dull edges, and load capacity.

Language-conditioned robotics. Recent systems use language models, multimodal models, or robot foundation models to guide skill choice and manipulation (Ahn et al., 2022; Huang et al., 2022a; Liang et al., 2023; Huang et al., 2023; Driess et al., 2023; Brohan et al., 2022; 2023; Open X-Embodiment Collaboration, 2023; Shridhar et al., 2022). Other work highlights that language models can struggle with planning (Huang et al., 2022b; Valmeekam et al., 2023). Our hostile distribution decouples object names from causal affordance variables. In that setting, text priors are the object of critique, not the mechanism.

Interactive perception, self-supervision, and world models. Robots can learn by poking, grasping, or predicting future observations (Agrawal et al., 2016; Pinto & Gupta, 2016; Ha & Schmidhuber, 2018). EATL can use such models as implementations of tests. The new claim is narrower: the meaning of a robot common-sense predicate is the executable evidence that certifies it for the current robot-object-demand tuple.

3 EXECUTABLE AFFORDANCE TEST LOGIC

Let o be an object instance, r a robot embodiment, a a candidate action family, and d a task demand such as volume, load, temperature, or required precision. A conventional prior estimates $p(y \mid \ell, x)$ where ℓ is the object label and x is a passive observation. EATL instead attaches a test program to each predicate.

Definition 1 (Executable affordance test). *An executable affordance test for predicate ϕ is a tuple $T_\phi = (\pi_\phi, g_\phi, c_\phi)$. The micro-policy π_ϕ chooses a bounded diagnostic action for (r, o, a, d) , the observation function returns a trace τ , and the guard $g_\phi(\tau)$ returns PASS, FAIL, or ABSTAIN. The scalar c_ϕ is the normalized test cost.*

Definition 2 (EATL witness). *The robot may assert $\phi(r, o, a, d)$ only if it has a witness (T_ϕ, τ) such that $g_\phi(\tau) = \text{PASS}$ under the current robot, object, action, and demand. If the guard returns FAIL, the trace is a failure certificate; if it returns ABSTAIN, the predicate is unavailable for safe planning.*

Examples are deliberately mundane. `can-contain-liquid` is tested by a safe pour probe that observes leakage and stability. `can-support-load` places a calibration load below the task demand. `can-cut-soft-food` uses a low-force edge probe on a sacrificial material.

can-wipe-spill measures absorbed volume. can-carry-hot tests heat resistance and handle availability. These are not full task rollouts. They are cheap pre-task experiments whose observations target the causal variables that a name or passive image may not expose.

4 A SIMPLE LOWER BOUND

The formal point is intentionally small: if the only input is a label and the deployment process can preserve that label while flipping the affordance, no text prior can recover the hidden variable.

Proposition 1 (Label-only error under affordance flips). *Fix an object label ℓ and binary affordance predicate Y . Any classifier h that depends only on ℓ has conditional error*

$$\Pr[h(\ell) \neq Y \mid L = \ell] \geq \min(p_\ell, 1 - p_\ell),$$

where $p_\ell = \Pr[Y = 1 \mid L = \ell]$. If a deployment distribution presents balanced same-label counterexamples, $p_\ell = 1/2$, the error is at least $1/2$. If an executable test observes sufficient evidence S with guard $g(S)$ satisfying $\Pr[g(S) \neq Y] \leq \epsilon$, then the test-based predicate has error at most ϵ on non-abstaining trials.

Proof sketch. For a fixed label, $h(\ell)$ is a constant. If it predicts 1, its error is $1 - p_\ell$; if it predicts 0, its error is p_ℓ . The best label-only choice therefore has error $\min(p_\ell, 1 - p_\ell)$. The test claim follows directly from the assumed guard error. The result does not say EATL tests are free or automatically available; it says the missing input channel is a real failure mode for label-only common sense.

5 SYNTHETIC STRESS TEST

We implement a transparent tabletop world with ten named object categories: mug, bowl, sieve, knife, spoon, sponge, brick, cardboard box, plastic tray, and glass. Each instance has hidden physical variables including concavity, porosity, holes, capacity, mass, friction, load capacity, sharpness, edge stiffness, absorbency, heat resistance, handles, stability, and flatness. Ground-truth affordances are deterministic functions of these variables and the task demand.

We compare three mechanisms:

- **text.prior**: a hand-coded name prior such as mugs contain liquid and knives cut.
- **vision.proxy**: a passive geometry proxy that sees shape-like variables but misses hidden material, damage, and wear.
- **eatl**: executable tests that probe the task-specific causal variables with small independent observation noise.

The regimes are: typical objects with mild wear; label-preserving flips where names stay fixed while hidden physical variables change; label swaps where an object inherits another object’s physical variables but keeps its name; and a mixed regime. The run uses 1000 episodes and 180000 method-task rows.

Table 1 shows the intended failure. The text prior is reasonable on typical objects, but under label-preserving flips it has 0.134 unsafe false positives. The vision proxy has even higher unsafe false positives because visible geometry often remains plausible while material or damage state changes. EATL reduces unsafe false positives to 0.022 by testing the deciding variable. Its mean normalized test cost is 0.095, representing the overhead of the diagnostic micro-experiment.

6 DISCUSSION

The result changes the central mechanism. If common sense is a prior, then the natural repair is a larger model, more data, uncertainty, or a verifier. If common sense is executable, the repair is different: define the predicate by the physical evidence that would make it true for this robot and this demand. That is why the proposed mechanism is not a new benchmark alone. The hostile benchmark exists to expose the assumption, but the contribution is the operational semantics.

Regime	Method	Acc.	Unsafe FP	Avoidable FN	F1
typical	text_prior	0.917	0.029	0.054	0.871
typical	vision_proxy	0.850	0.142	0.007	0.813
typical	eatl	0.980	0.014	0.006	0.971
label preserving flips	text_prior	0.834	0.134	0.033	0.680
label preserving flips	vision_proxy	0.787	0.207	0.006	0.656
label preserving flips	eatl	0.972	0.022	0.007	0.934
label swap	text_prior	0.679	0.152	0.169	0.500
label swap	vision_proxy	0.840	0.150	0.010	0.800
label swap	eatl	0.971	0.020	0.009	0.956
mixed	text_prior	0.797	0.113	0.089	0.668
mixed	vision_proxy	0.819	0.170	0.011	0.758
mixed	eatl	0.956	0.031	0.013	0.928

Table 1: Runnable synthetic stress test. Unsafe false positives are cases in which a method asserts an affordance that is false, e.g., approving a leaking container or overloaded support.

EATL also changes how failures are represented. A failed pour probe is not just a low confidence score; it is a certificate that the current object does not satisfy the containment predicate for the demanded volume. This is useful for planners because the failure is local, typed, and reusable until the object state changes.

7 LIMITATIONS

This paper does not show real-robot performance. The tests are hand-specified, the cost model is simplified, and the synthetic environment assumes the test observes sufficient causal variables with independent noise. Some predicates may require destructive tests, long-horizon interaction, or specialized sensors. Passive perception and large multimodal models can also solve many cases when the relevant variable is visible. The claim is therefore not that EATL replaces perception, planning, or learned affordance models. The claim is that an embodied common-sense assertion should be treated as unsupported until the robot has executable evidence for the current physical relation.

8 CONCLUSION

Robots fail when familiar words hide changed physics. We proposed EATL, a semantics in which common-sense affordance predicates are certified by cheap robot-specific tests rather than inherited from text priors. A simple lower bound shows why label-only priors cannot solve label-preserving affordance flips, and a runnable synthetic stress test demonstrates the safety effect on containment, lifting, support, cutting, wiping, and hot carrying. The next step is to learn or synthesize test programs and evaluate them on real robot manipulation tasks with explicit test cost and failure risk.

REPRODUCIBILITY STATEMENT

The repository contains the literature builder, synthetic environment, episode logs, summary CSV, and table-generation script. The main experiment is reproduced with:

```
python experiments/affordance_tests.py
python scripts/write_paper_assets.py
```

The literature sweep is reproduced with `python scripts/build_literature.py`; OpenAlex rate limiting during this run was recovered using Crossref and arXiv metadata, as recorded in `data/literature_progress.json`.

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